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*36TH WING*

*DEPARTMENT OF THE AIR FORCE*
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*Maintenance*

# *AIRCRAFT AND EQUIPMENT MAINTENANCE MANAGEMENT*

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***SUMMARY OF CHANGES***

This document has been substantially revised and must be completely reviewed. Major changes include the addition of several paragraphs, several paragraphs were realigned and renumbered to match the current DAFI21-101 and DAFI21-101_PACAFSUP, paragraphs 15.2.6.1.2 – 15.2.6.1.5 & **Figure 15.1**. – **Figure 15.2** were deleted, **Attachment 11** and **Attachment 12** changed and combined to **Attachment 14** with revised *Hangar Entry Checklist*, Attachment 13 changed to **Attachment 15**, **Attachment 14** changed to **Attachment 16** and **Attachment 17** **added** *36 WG AF145, LOST TOOL/OBJECT REPORT*. Information in parent regulations, repeated information and extraneous information has been removed from this supplement.

**1.15. Communications.**

1.15.2.1. **(Added-36WG)** Personnel will also comply with restrictions in TO 00-25-172, DAFMAN 91-201 *Explosive Safety Standards*; DAFI 91-207, *The Traffic Safety Program*; AFI 31-218, *Motor Vehicle Traffic Supervision*; DAFMAN 17-1301, *Computer Security (COMSEC)* and aircraft and equipment TOs.

1.15.2.2. **(Added-36WG)** Personal electronic and communication (commercial mobile) device restrictions vary by weapon system and location. It is each individual's responsibility to know requirements and account for their security.

1.15.2.3. **(Added-36WG)** Cell phone cameras are to be used for official use only, and only by persons authorized on a Flightline Photo Authorization Letter and/or Controlled Area Photo Authorization Letter for the 36th Munitions Squadron (MUNS) Munitions Storage Area(s) (MSA).

**2.4. Maintenance Group Commander (MXG/CC) Responsibilities.**

2.4.1.2.1. **(Added-36WG)** Refer to **Attachment 15** for Radiation Protection Program guidance.

2.4.2.14.1. **(Added-36WG)** The 36 MXG/CC's Functional Check Flight (FCF), Operational Check Flight (OCF), and High-Speed Taxi Check policy: Integrated/deployed units will use home station developed procedures in coordination with 36 OG/CC.

2.4.3.3.1. **(Added-36WG)** The 36 MXG/CC's Lead the Fleet (Pacer) Program policy: Integrated/deployed units will use home station procedures.

2.4.3.14.1. **(Added-36WG)** An appointed 36 MXG permanent party Quality Assurance Evaluator will assist integrated/deployed personnel to implement the 36 MXG Impoundment Program procedures. The Quality Assurance office is the OPR for the Impoundment Program.

2.4.3.15.1. **(Added-36WG)** 36 MXG/CC's policy to review and clear "repeat", "recur", and "cannot duplicate" (CND) discrepancies: Integrated/deployed units will use home station procedures.

2.4.4.1.1.1. **(Added-36WG)** 36 MXG/CC's Flying Crew Chief (FCC) policy: Integrated/deployed units will use home station unit program(s).

**3.1. (36WG) General.** Andersen Air Force Base (AAFB) does not have an AMXS due to there being no permanently assigned aircraft. However, the 36 EAMXS will help support the deployed/TDY units with accomplishing their mission. The deployed/TDY's home station MXG/CC or designated representative will be the authorizing official to sign off/clear owning aircraft and equipment discrepancies. Furthermore, all incidents happening at AAFB will be reported to 36 MXG/MOC or TDY Units' MOC representative for QRC/notification purposes (e.g., CANN authorizations, OTF, and lost items).

**4.4. Accessories Flight.**

4.4.4.1.5. **(Added-36WG)** Deployed/TDY Fuel System personnel shall maintain and account for their external fuel tanks, conformal fuel tanks (CFTs), and weapons bay tanks (WBTs). All tanks shall be completely drained of any fuel prior to removal from aircraft. All tanks will be stored on manufactured or locally made stands IAW applicable technical data. Follow aircraft specific safety guidelines when performing maintenance on tanks.

4.4.4.2.1.1. **(Added-36WG)** All fuel cell maintenance requiring tank entry shall be coordinated with the 36 MXS Pro Super and the Andersen AFB Fire Department.

4.4.4.2.1.2. **(Added-36WG)** Primary open fuel system repair areas are listed on the current Confined Space Master Entry Plan. Please see the most current version provided by QA for a list of approved locations.

4.4.4.2.1.3. **(Added-36WG)** The work site supervisor will notify the 36 MXS Production Supervisor and then notify proper authorities before fuel system maintenance begins and after the open tank/open fuel source is closed IAW Technical Order 1-1-3, *Inspection and Repair of Aircraft Integral Tanks and Fuel Cells*.

4.4.4.2.1.4. **(Added-36WG)** Deployed/TDY units will follow the Andersen AFB Aircraft Confined Space Master Entry Plan IAW Technical Order 1-1-3, *Inspection and Repair of Aircraft Integral Tanks and Fuel Cells*.

**4.5. Aerospace Ground Equipment (AGE) Flight.**

4.5.1.4.1. **(Added-36WG)** Perform chassis, enclosure, and trailer maintenance on fuel bowsers. The Unit/Flight that fills fuel bowsers with product are responsible for transporting, funding contract costs for emptying, and returning fuel bowsers to the AGE sub pools and/or AGE facility (Bldg. 18004).

**4.8. Fabrication Flight.**

4.8.3.7. **(Added-36WG)** Welding operations shall be performed indoors in Building 18006 annex with properly rated fire extinguisher and fireguard. When an item requires welding and cannot be brought to the Fabrication Flight, a DAF 592 will be coordinated through the 36th Wing Safety office.

**6.15. Weight and Balance (W&B) Program.**

6.15.2.1. **(Added-36WG)** Weight and Balance (W&B) Program. Units will utilize their home station guidance and checklists for completing W&B on assigned aircraft. Deployed QA W&B qualified technicians will hand carry all W&B documentation for their units assigned aircraft to home station upon completion of deployment.

**7.3. Impoundment Authorities.**

7.3.1.1.1. **(Added-36WG)** If the impoundment is due to Foreign Object Damage (FOD) and/or Dropped Object Prevention (DOP) incidents, notify the 36 WG/FOD monitor, and 36 WG/SE. Forward a copy of the report to the 36 WG FOD/DOP monitor.

**7.5. Mandatory Impoundments.**

7.5.12. **(Added-36WG)** If the impoundment is due to Foreign Object Damage (FOD) and/or Dropped Object Prevention (DOP) incidents, notify the 36 WG/FOD monitor, and 36 WG/SE. Forward a copy of the report to the 36 WG FOD/DOP monitor.

7.5.13. **(Added-36WG)** Impoundment is mandatory if an aircraft is suspected of utilizing AGE and fuel-related equipment that may have been utilized on a separate mishap aircraft.

**8.1. Tool and Equipment Management.**

8.1.1. **(Added-36WG)** Munitions Squadrons/Flights will manage tools and equipment as established in DAFMAN 21-200, Chapter 9.

8.1.2. **(Added-36WG)** Tool and equipment management will be accomplished in conjunction with DAFI 21-101, DAFI 21-101_PACAFUP, and this supplement.

**8.2. Guidelines for Program Management.**

8.2.2.1. **(Added-36WG)** All tools will be inventoried and documented annually. The most current inventory will be signed by the CTK Custodian and kept in TAB F of the Master Composite Tool Kit (CTK) Continuity Book.

8.2.2.2. **(Added-36WG)** CTK Custodians will ensure all tools are inspected for serviceability IAW TO 32-1-101, *Use and Care of Hand Tools and Measuring Tools*, and owner's manuals.

8.2.2.3. **(Added-36WG)** CTK Custodians will document and sign spare tool inventories and maintain the most current inventory in TAB F in their Master CTK Continuity Book.

8.2.4.1. **(Added-36WG)** With the exception of spare HAZMAT, all spare tools/consumables will be stored in a lockable container that is physically separated from issuable tools/equipment if space and funds allow.

8.2.4.2. **(Added-36WG)** HAZMAT approved storage cabinets will be clean and free from exposed HAZMAT and FOD.

8.2.4.3. **(Added-36WG)** To prevent accidental/unprotected exposure to HAZMAT, containers will not have excess accumulation or pooling of HAZMAT on the outside of the container when stored in an approved HAZMAT cabinet/location (i.e. brake fluid on top of container around lid, pooling of HAZMAT on a 50-gallon drum lid).

8.2.4.4. **(Added-36WG)** The bottom of the HAZMAT storage cabinets will not be used as a shelf and will be kept empty at all times. The bottom is designed to prevent HAZMAT spills from leaking out of the cabinet.

8.2.5.1.1. **(Added-36WG)** CTK turnover at the jobsite is authorized on a limited case-by-case basis as determined by the flight chief, production superintendent, or shift supervisor. A 36MXG Form 140, *CTK Inventory and Control Log*, must be utilized to document the inventory.

8.2.6.1. **(Added-36WG)** For Munitions Squadrons/Flights, lost tool procedures are further defined in DAFMAN 21-200.

8.2.9.4. **(Added-36WG)** Rag control applies to all 36 WG maintenance organizations to include units that are deployed into MXG. Rag control requirements are as follows:

8.2.9.4.1. **(Added-36WG)** Spare rags must be positively controlled. Document completed inventory of rags at the beginning and end of each shift in TCMax®.

8.2.9.4.2. **(Added-36WG)** Issue a predetermined number of rags in TCMax® (example; assign 5 each to TCMax® ID). When rags are combined with CTKs, include them on the Master Inventory List (MIL). When rags are issued in pre-packaged containers, mark each container with the number of rags. All rags issued for deployments or TDYs must be included during inventories and all local policies must address long-term issuance.

8.2.9.4.3. **(Added-36WG)** All replacement rags must be issued on a one-for-one swap.

8.2.9.4.4. **(Added-36WG)** Dirty and clean rags must be segregated.

8.2.11.1. **(Added-36WG)** Locally developed/manufactured tools shall be added to the CTK inventory, assigned a CTK identification number and issued/controlled in the same way as all other tools/equipment.

8.2.12.1. **(Added-36WG)** Agencies not associated with 36 MXG (i.e., depot teams, factory representatives, contract field teams, etc.) working on equipment within the unit may use assigned CTK(s) and test equipment. A member of the 36 MXG shall complete inventory with the outside agency member upon CTK/equipment initial issue and final return.

8.2.13.2. **(Added-36WG)** A physical copy of the current MIL will be maintained with the decentralized CTK.

8.2.14.2. **(Added-36WG)** Equipment and tools in emergency response vehicles, trailers, and transportable maintenance systems will be considered a dispatchable CTK. They will have a 9-digit CTK number assigned and follow dispatchable CTK guidance.

8.2.14.3. **(Added-36WG)** Positive control of equipment will be maintained in the same way as all other tools/equipment.

8.2.14.4. **(Added-36WG)** Keys for equipment/tools will remain with keys to the emergency response vehicle. All keys will be tracked and accounted for in TCMax®.

8.2.15.2. **(Added-36WG)** In the event that there is only one person in the work center to sign a tool kit in or out, the individual will request a second party noncommissioned officer (NCO), Production Supervisor, or shift supervisor to perform inventory/sign in the CTK.

8.2.16.1. **(Added-36WG)** Access to tool rooms will be limited to authorized personnel. The Section Chief or higher will sign a letter of personnel authorized unescorted access to the tool room.

**8.3. General Program Guidelines.**

8.3.1.2. **(Added-36WG)** Each CTK custodian shall maintain a Master CTK Continuity Book (physical or digital) containing the following tabs and applicable information:

8.3.1.2.1. **(Added-36WG)** Tab A: Appointment Letter(s) (Most Current)

8.3.1.2.2. **(Added-36WG)** Tab B: MAJCOM Guidance

8.3.1.2.3. **(Added-36WG)** Tab C: Local Guidance

8.3.1.2.4. **(Added-36WG)** Tab D: List of Assigned CTKs

8.3.1.2.5. **(Added-36WG)** Tab E: Master Inventory Listing (MIL) for Assigned CTKs

8.3.1.2.6. **(Added-36WG)** Tab F: Most Recent CTK Inspection Logs

8.3.1.2.7. **(Added-36WG)** Tab G: Locally Manufactured Tools and Equipment Listing (letters and drawings)

8.3.1.2.8. **(Added-36WG)** Tab H: Spare Tool Inventory Listing

8.3.1.2.9. **(Added-36WG)** Tab I: Miscellaneous Items

8.3.5.1.1. **(Added-36WG)** Tools/equipment which are permanently removed from a CTK shall have their inlays, silhouettes, or compartments filled in. Ensure filler does not pose a FOD hazard.

8.3.6.4.3. **(Added-36WG)** When practical, dispatchable CTKs shall contain a FOD pouch/bag with contrasting 2-inch letters, at a minimum, depicting "FOD" stenciled on it. Track the pouch/bag on the MIL and empty it when full and at the end of each shift.

8.3.6.5.2. **(Added-36WG)** Consumables and expendable hand tools (i.e., blades, apexes, file cleaners, etc.) that are exhausted through use shall be replaced on a one-for-one basis. Annotate on the MIL and update TCMax® when items are removed from tools/equipment.

8.3.6.6.2. **(Added-36WG)** The container will count as one piece unless the container is required to be separated for use. If the container is required to be separated for use, the MIL will be annotated to identify the total number of pieces in the tools kit plus the total pieces of the container. For example, a test set might show 7 each pieces + 4 each container pieces for an overall total of 11 each.

8.3.6.7.5. **(Added-36WG)** Unserviceable CTKs, tool bags, toolboxes, cases, tools, equipment, items, and test sets will be removed from service. They will be segregated, clearly marked as unserviceable (i.e. AFTO Form 350, DD Form 1500 series tag, tape…), and annotated in TCMax® and on the MIL. This applies only to the above that are awaiting repair or awaiting parts and are expected to be returned to service.

8.3.6.7.5.1. **(Added-36WG)** Unserviceable CTKs, tool bags, toolboxes, cases, tools, equipment, items, and test sets that are not awaiting repair/parts will be permanently removed from service. Permanent removal will be annotated in TCMax® and on the MIL.

8.3.6.7.5.2. **(Added-36WG)** Broken/Removed tools under warranty shall be segregated, inventoried, and secured until turned in for replacement.

8.3.6.7.5.3. **(Added-36WG)** Unserviceable tool bags, toolboxes, cases, tools, equipment, items, and test sets will not be stored in CTKs.

8.3.6.7.5.4. **(Added-36WG)** Any Chipped or broken piece(s) that does not affect the serviceability of the tool, will be annotated in TCMax® and on the MIL.

8.3.6.8. **(Added-36WG)** Dispatchable support equipment (SE) or special tools (ST) inventory must match the MIL in TCMax®. The dispatchable SE/ST will follow the same rules for broken/missing items as outlined in **paragraph 8.9** of this supplement.

8.3.14. **(Added-36WG)** All CTKs, test equipment and storage containers that are subject to flight line use will be outlined using reflective surface on each edge.

**8.5. Tool Accountability.**

8.5.1.2.1.2. **(Added-36WG)** All assigned tools, equipment, tool kits, Hazardous Material (HAZMAT) items and Technical Orders (TOs) shall be entered in TCMax® or contractor database using an assigned 9-digit Worldwide Identification (WWID). Items shall be issued in TCMax® when removed from the tool room/assigned location.

8.5.1.2.1.3. **(Added-36WG)** HAZMAT quantities, issuance, and receipt shall be tracked to ensure accountability in the same manner as tools in TCMax® or a MAJCOM-approved accountability system using a WWID format. Each HAZMAT item shall have a label or be marked with the WWID. Multiple items with the same national stock number can be used as one WWID. HAZMAT items installed or attached to machinery shall be tracked in TCMax® as consumed/expended (i.e., attached tank of acetylene that goes to welding equipment).

8.5.1.2.7. **(Added-36WG)** Track and manage Test Measurement and Diagnostic Equipment (TMDE) calibration dates in TCMax®.

8.5.1.2.8. **(Added-36WG)** Identify storage locations for all decentralized items, items not secured in the tool/support room, that are required to be tracked in TCMax®.

8.5.1.4. **(Added-36WG)** All assigned non-dispatchable TOs, equipment, and HAZMAT shall be tracked using the AF Form 1297, *Temporary Issue Receipt*, or 36WG Form 140, *CTK Inventory and Control Log,* for accountability and control when removed from the tool room/assigned location. TCMax® may be utilized as an alternate tracking system.

**8.6. Tool and Equipment Marking and Identification.**

8.6.1.1.3. **(Added-36WG)** Certain tool kits are sold as a whole and not by individual pieces. To prevent unnecessary work delays, a stock of spare tool kits (e.g. Allen wrench sets and drill bit sets) are authorized. Kits stored in spare tools are normally used to replace a complete kit set. However, if a specific individual item is required for a replacement and is removed from a kit set, the removed item will be documented in TCMax® and on the kit itself. Documentation for the kit in TCMax® and on the kit itself will consist of the total amount of pieces in the kit to include the case (e.g., 7 pcs + case = 8 total).

8.6.1.1.4. **(Added-36WG)** With the exception of spare HAZMAT, all spare tools/consumables will be stored in a lockable container that is physically separated from issuable tools/equipment if space and funds allow.

8.6.1.3.1. **(Added-36WG)** All 36 WG CTKs, tools, TOs, equipment, and HAZMAT accessing the Flightline and/or FOD Critical areas, will use the following standards for the EID (tools located inside the toolbox need only the first 5 characters):

**Table 8.1. (Added-36WG) Standards for the EID.**

| POSITION 1-3 | POSITION 4 | POSITION 5-9 |
|---|---|---|
| **36th Maintenance Squadron**<br>ATX | Flight/Section Identifier: (Letter)<br>(A) = AGE MX Floor<br>(B) = Not used<br>(C) = AGE Support<br>(D) = Not used<br>(E) = Egress<br>(F) = E/E<br>(G) = Pneudraulics<br>(H) = Fuels<br>(I) = Not Used<br>(J) = Corrosion<br>(K) = NDI<br>(L) = Metals Technology<br>(M) = Structural<br>(N) = Transient Alert<br>(O) = MilSpray<br>(P) = Crash Recovery<br>(Q) = Not Used<br>(R) = Not Used<br>(W) = Armament<br>(S-V, X-Z) = Reserved | Any combination of numbers/letters |
| **36th Expeditionary Aircraft Maintenance Squadron**<br>ATA | Flight/Section Identifier: (Letter)<br>(B) = Bomber Mx<br>(M) = MOC/Wheel & Tire<br>(Q) = Quality Assurance | |
| **36th Munitions Squadron**<br>ATU | Flight/Section Identifier: (Letter)<br>(A) = Operations<br>(C) = Combat Munitions Training<br>(E) = Equipment Maintenance<br>(M) = PGM<br>(N) = Inspection<br>(P) = Munitions Maintenance<br>(S) = Storage<br>(T) = Control<br>(I) = Not Used<br>(B, D, F-L, O, Q, R, U-Z) = Reserved | |

| | | |
|---|---|---|
| **Det 1, 36th Mission Support Group, Diego Garcia**<br>ATD | Flight/Section Identifier: (Letter)<br>(A) = AGE<br>(M) = Munitions | Any combination of numbers/letters |
| **497 Combat Training Flight, Singapore**<br>ATC | Flight/Section Identifier: (Letter)<br>(S) = Support | |
| **36th Logistics Readiness Squadron, Fuels Management Flight**<br>ATL | Flight/Section Identifier: (Letter)<br>(F) = Fuels Management<br>(M) = Maintenance Section<br>(O) = Vehicle Maintenance<br>(V) = DZSP21 Vehicle Maintenance | |
| **36th Operations Support Squadron, Aircrew Flight Equipment Section**<br>ATOF | | |
| **36th Mobility Response Squadron, Aircraft Section**<br>ATR | | |
| **734th Air Mobility Squadron**<br>ATS | Flight/Section Identifier: (Letter)<br>(T)=Maintenance Section | |
| **4th Reconnaissance Squadron, Global Hawk**<br>ATG | | |
| **36th Civil Engineering Squadron**<br>ATE | Flight/Section Identifier: (Letter)<br>(B) = Entomology<br>(E) = Electric/Alarms<br>(F) = Fuels<br>(G) = Ground Maintenance<br>(H) = HVAC<br>(P) = Plumbers<br>(V) = Vertical Repair<br>(W) = Power Production<br>(Z) = Horizontal | |

| | | |
|---|---|---|
| **506th Expeditionary Air Refueling Squadron**<br>ATM | Flight/Section Identifier: (Letter)<br>(A) = Flightline<br>(C) = Inspection<br>(E) = Electrics<br>(F) = Fuels<br>(G) = AGE<br>(H) = Hydraulics<br>(M) = Metals<br>(N) = NDI<br>(P) = Prop<br>(R) = R&R<br>(S) = Structural<br>(V) = Avionics | |
| **Helicopter Sea Combat Squadron 25**<br>QQ0 | | |
| **36th Contingency Response Squadron**<br>(ATB) | Flight/Section Identifier: (Letter) | |

8.6.1.4.6. **(Added-36WG)** HAZMAT items will be marked with a 9-digit EID. The EID must tie the HAZMAT item back to its designated storage location. HAZMAT items issued for one time use (oil cans, hydraulic cans, mixing compounds, etc.) do not require a WWID marking.

8.6.2.4. **(Added-36WG)** The containers must also be marked with a 9-digit EID. The EID must tie the container back to its designated storage location.

8.6.8. **(Added-36WG)** All containers/tool kits will be marked with the same EID as the item(s) contained inside (i.e. torque wrench case, drill bit case…).

**8.7. Locally Manufactured, Developed, or Modified Tools and Equipment.**

8.7.4. **(Added-36WG)** All unit representatives who have locally manufactured tools/equipment that require approval shall first contact the 36 MXG/QA Locally Manufactured Tool Program Monitor. The program monitor will instruct the technician on current procedures (e.g., approval letter format, routing requirements, and photographs/drawings).

8.7.4.1. **(Added-36WG)** Locally developed/manufactured tools shall be issued/controlled in the same manner as all other tools/equipment.

8.7.4.2. **(Added-36WG)** Local manufacturing procedures are as follows:

8.7.4.2.1. **(Added-36WG)** The Fabrication Flight Chief initially approves or disapproves all requests for locally manufactured parts/tools for 36 MXG units, while the final approving authority is the 36 MXG/CC or designated representative. Deployed/TDY units will follow their home station procedures.

8.7.4.2.2. **(Added-36WG)** Manufacture of procurable items (those not source code JBD) is restricted to those which are mission-essential. Manufacture of items Coded JBB are also limited to mission needs. For local manufacture, mission-essential items must include MICAP documentation. For items not MICAP but impacted by extended estimated delivery dates (EDDs), coordination is required through the 36 MXS Production Superintendent. If approved, the Pro Super will sign the designated acceptance line on the request form, acknowledging that the request is properly documented, recommending fabrication of the item while awaiting delivery of the ordered part.

8.7.4.3. **(Added-36WG)** Local manufacture requester must:

8.7.4.3.1. **(Added-36WG)** Submit an MXG Locally Manufacture Tool Approval Memorandum to the 36 MXS Fabrication Flight (MXMF). Contact the 36 MXG/QA Locally Manufactured Tool Program Manager to receive a copy of the memorandum.

8.7.4.3.2. **(Added-36WG)** Submit engineering drawings or any applicable technical order data that is relevant to the manufacturing process.

8.7.4.3.3. **(Added-36WG)** The requester will fill out a DD Form 1348-6, if deemed necessary by the fabricating section for reimbursement of supplies and additional costs.

8.7.4.3.4. **(Added-36WG)** The requester, in conjunction with the appropriate fabricating section shall determine all required materials, tooling, and consumables necessary to support the manufacturing process. The requestor retains full responsibility for ordering all required items, including coordination with Supply personnel or their designated Resource Advisor. Additionally, the requestor will maintain custody of all parts and materials until all components are received and ready to proceed with the manufacturing of the item.

8.7.4.3.5. **(Added-36WG)** Locally manufactured items that require routing through multiple sections in the Fabrication Flight shall be coordinated before the job can begin. All sections shall be listed on individual AFTO Form 350 tags.

**8.8. Tool Room Operations and Security.**

8.8.2.1.3. **(Added-36WG)** Tool rooms left unattended by appointed authorized personnel, will be locked to prevent unauthorized access and maintain positive tool control and security.

**8.9. Lost Item/Tool Procedures.**

8.9.3. **(Added-36WG)** All units will refer to 36WGI21-110, *Foreign Object Damage (FOD), Dropped Object Prevention (DOP), Program, and Tool Accountability*, (unless covered by DAFMAN 21-200 para 9.6.2.1.1.) for proper lost tool/object procedures and for further guidance. This covers all lost tools, objects, or equipment regardless of being dispatchable, non-dispatchable, or the potential for FOD in the flightline. This information will be tracked by 36 MXG QA for tool accountability/trend analysis who will then report it to the 36 WG FOD/DOP monitor.

**9.5. Bench Stock.**

9.5.3.3. **(Added-36WG)** Ensure Op Stock, Bench Stock, and Shop Stock item segregation is maintained at all times. Items will only be stored in their assigned bins. Co-mingling is not authorized to ensure proper AF resource stewardship.

**11.6. Red Ball Maintenance.**

11.6.5.2. **(Added-36WG)** Deployed/TDY units will use home station Red Ball procedures.

**11.8. Foreign Object Damage (FOD) Prevention Program.**

11.8.3.7.2. **(Added-36WG)** All equipment added to a vehicle for daily use, to include a FOD pick tool, FOD container, flashlight, WEX cards and/or VIL Key (If applicable) as a minimum, shall be marked/etched or stamped with the assigned vehicle registration number or EID assigned to the applicable vehicle. To provide accountability and prevent FOD, the aforementioned items, and any additional items, must be individually annotated on the back of the vehicle's AF Form 1800, *Operator's Inspection Guide and Trouble Report* (General Purpose Vehicles).

**11.9. Dropped Object Prevention (DOP) Program.**

11.9.3.1.3. **(Added-36WG)** Deployed units shall send reports to the 36 WG/DOP Monitor for trending purposes and forward a copy of the report to home station.

**11.12. Radar Warning Receiver/Radar Threat Warning Testing.**

11.12.2.2.2. **(Added-36WG)** Deployed/TDY Avionics managers will be appointed as the Radar Warning Receiver (RWR)/Radar Threat Warning (RTHW) Manager by their home station MXG/CC. RWR/RTHW will be checked on all applicable aircraft per home station requirements. When an aircraft is found to have a malfunctioning RWR/RTHW system, the Aircrew and Deployed/TDY unit Production Superintendent determines the course of action based on operational needs and requirements.

**11.42. (Added-36WG) Aircraft Hangar Entry.**

11.42.1. **(Added-36WG)** Aircraft Hangar Entry Checklist and Brief, and applicable MDS technical data shall be used prior to towing/positioning aircraft in the hangar respectively. See **Attachment 14**.

11.42.2. **(Added-36WG)** The aircraft-towing supervisor will use and annotate a hangar entry/removal checklist on each aircraft in the hangar before towing operations begin. Checklists must be kept with aircraft forms while in the hangar. The towing supervisor will initial each item separately, ensuring the checklist is accomplished and the aircraft is properly configured for hangar entry/removal. The towing supervisor will print name (Rank, Last, First Name) and sign the hangar entry/removal checklist.

**11.43. (Added-36WG) Bomblift Control Procedures.**

11.43.1. **(Added-36WG)** Bomblifts are controlled assets and will be signed out using an AF FORM 1297, *Temporary Issue Receipt*, from the respective AGE team. Only bomblifts with current service inspections will be issued out to operators. The AGE representative will check the bomblift forms for status prior to issuing. Units requesting bomblifts will ensure their personnel are qualified to operate bomblifts. If personnel are not qualified/certified to operate a bomblift, personnel will not be authorized to take possession of the bomblift. An operator inspection will be conducted by the operator prior to starting and taking custodial responsibility of the bomblift per TO 00-20-1, *Aerospace Equipment Maintenance Inspection, Documentation, Policy, and Procedures*.

11.43.2. **(Added-36WG)** Issued bomblifts will be returned to the AGE facility, Bldg. 18004, by the operator who signed out the bomblift or the operator's assigned unit at 7-day intervals to ensure serviceability. Bomblift service inspections will be performed and documented monthly with the exception of assets placed in WRM storage on part II of the DPAS/AFTO Form 244 by 36 MXS/AGE personnel only. If a service inspection is not conducted by 36 MXS/AGE personnel within 90 days, that equipment will be deemed unserviceable on the DPAS/AFTO Form 244 and removed from all locations, with the exception of the AGE facility, Bldg. 18004, until an inspection can be accomplished, deemed serviceable, and properly documented.

**11.44. (Added-36WG) Inspection and Operation of AGE.**

11.44.1. **(Added-36WG)** All qualified/certified users of AGE shall perform an Operator Inspection prior to starting and taking custodial responsibility. This inspection is intended to ensure serviceability and safety of the equipment prior to use. It consists of a review of the forms (paper or electronic) for current status, and a visual inspection of the equipment for defects and adequate servicing. The operator inspection is the responsibility of the user. The operator inspection shall be documented in part II of the DPAS/AFTO Form 244.

11.44.2. **(Added-36WG)** All organizations using AGE will have qualified operators to monitor and ensure safe operation of the equipment. Qualification training may be coordinated through 36 MXS AGE supervision.

11.44.3. **(Added-36WG)** Damaged/abused equipment and lost/missing hardware will be reported to the applicable AGE team immediately upon discovery. The individual discovering damaged/abused equipment and lost/missing hardware will enter the appropriate symbol and discrepancy on the equipment DPAS/AFTO Form 244. Additionally, the individual discovering damaged/abused equipment or lost/missing hardware will initiate an incident report along with the 36MXG Form 145, *Lost Tool/Object Report*.

CHARLES D. COOLEY
Brigadier General, USAF
Commander

**Attachment 1**

**GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION**

***References***

None

***Prescribed Forms***

**(Added-36WG)** 36 MXG Form 140, *CTK Inventory and Control Log*

**(Added-36WG)** 36 WG AF145, *Lost Tool/Object Report*

***Adopted Forms***

DAF Form 847, *Recommendation for Change of Publication*

***Terms***

**(Added-36WG) Dispatchable**—Designed to be used outside of workcenter.

**(Added-36WG) Non-dispatchable**—Designed not to be used outside of workcenter.

**(Added-36WG) Work center**—Area within the organization's building.

## Attachment 14

## AIRCRAFT HANGAR ENTRY CHECKLIST

**Table A14.1. 36 MXG Aircraft Hangar Entry Checklist.**

| ALL PURPOSE CHECKLIST | | 36th MOC: 366-4321 | | |
|---|---|---|---|---|
| TITLE/SUBJECT/ACTIVITY/FUNCTIONAL AREA<br>36th MXG Hangar Entry Checklist | | OPR<br>36 MXG/MXGQ | DATE | |
| ***Aircraft Serial Number:*** | ***Owning Unit:*** | YES | NO | N/A |
| NOTE: The Tow Supervisor/Crew Chief is responsible for ensuring this checklist is complete and placed on the aircraft.<br><br>Warning! Hangar fire suppression systems are disabled. Emergency switches will only notify emergency services of an emergency. Fire bottles will need to be utilized in the event of a fire.<br><br>WARNING! Operating -95 Gas Turbine Compressors or -60 Gas Turbine Compressors inside of Hangars WILL activate the fire alarm system. | | | | |
| 1. Prior to use inspection on all hangar safety equipment (eye wash stations, fire extinguishers, etc.) must be complied with prior to entry. | | ☐ | ☐ | ☐ |
| 2. Actions Prior to Hangar Entry.<br>a. Weapons:<br>Remove all impulse carts, chaff/flare, munitions, and internal ammunition. | | ☐ | ☐ | ☐ |
| b. Crew Chiefs:<br>All safety pins/protective covers & -21 equipment installed. | | ☐ | ☐ | ☐ |
| Aircraft battery disconnected/capped (28V/270V, H25C for fuel cell maintenance or applicable maintenance) | | ☐ | ☐ | ☐ |
| c. Ensure all hangar doors are fully opened (including tail door if applicable). | | ☐ | ☐ | ☐ |
| 3. Hangar Entry Accomplished.<br>a. Aircraft properly chocked (chocks laced). | | ☐ | ☐ | ☐ |
| b. Aircraft properly grounded (with serviceable grounding wire). | | ☐ | ☐ | ☐ |
| c. WARNING/CAUTION cones displayed as required. (Mandatory if aircraft protrudes from hangar) | | ☐ | ☐ | ☐ |
| d. Ensure protective covers are installed (as applicable). | | ☐ | ☐ | ☐ |
| e. Position drip pans with absorbent pads under aircraft to contain fluid leaks. | | ☐ | ☐ | ☐ |
| f. Ensure serviceable fire bottles are in place as required, free of obstructions and fire lanes are clear. | | ☐ | ☐ | ☐ |
| g. Ensure emergency tow bar remains with aircraft. | | ☐ | ☐ | ☐ |
| Printed Name: | Signature: | | | |
| AF IMT 2519, 19911101 V5 | PREVIOUS EDITIONS ARE OBSOLETE. | | | |

## Attachment 15

## RADIATION PROTECTION PROGRAM POLICY

**A15.1. DAFI 21-101, paragraph 2.4.1.1., states** "Establish a radiation protection program IAW AFI 48-109, *Electromagnetic Field Radiation (EMFR) Occupational & Environmental Health Program*, when applicable. (T-1)." The paragraphs below are the required procedures.

**A15.2. Radiation Protection Program.** RF hazard zone recognition is the most important step in avoiding overexposure. Supervisors will ensure all personnel are familiar and comply with procedures outlined in this supplement and AFI 48-109, *Electromagnetic Field Radiation (EMFR) Occupational & Environmental Health Program*. When situations or conditions conflict with the following, deployed units must utilize their home station program procedures. The policy will, at a minimum, include:

A15.2.1. General Radiation Protection Procedures.

A15.2.2. Maintain RF hazard distances listed in the system specific technical orders when operating RF systems.

**A15.3. Specialists will:**

A15.3.1. Identify the perimeter of respective RF hazard zones with RF hazard warning signs or traffic cones. Exception: RF hazard warning signs will not be placed on the surface of the aircraft to cordon off radiating upper antennas. (A ground observer will be used to prevent personnel from entering RF Hazard Area).

A15.3.2. Signs must be marked and positioned IAW AFI 48-109, *Electromagnetic Field Radiation (EMFR) Occupational & Environmental Health Program*.

A15.3.3. RF radiation warning signs and cones will be posted in the upright position when systems are transmitting. When transmitting operations are complete, they will be laid flat on the ground or removed.

A15.3.4. Notify all personnel working on the aircraft (or in the area) when RF transmitting operations begin and brief the hazard zones.

A15.3.5. Deactivate the fire suppression system during system operations when radiating Offensive Radar or HF radio in the maintenance docks or hangars.

A15.3.6. Coordinate with the production superintendent and Owning Units' Designated MOC to establish the aircraft radar hazard zone.

A15.3.7. Full scan radiation is prohibited.

A15.3.8. Maintenance dock/hangar doors must be fully open (cones or warning signs will be used to cordon off area in front of dock/hangars).

A15.3.9. Only essential personnel will be inside the bay area during transmission and must remain outside or AFT of the AF radiation zone.

A15.3.10. Entrance doors will be posted with an RF hazard warning sign.

**A15.4. MOC representatives will:**

A15.4.1. Monitor the RF operations, assist the production superintendent with personnel notification and any other coordination as required.

**A15.5. Suspected Overexposures:**

A15.5.1. When personnel suspect they have been overexposed to radiation, the production superintendent will immediately notify the corresponding MOC and ensure the individual(s) is transported to the hospital.

A15.5.2. Notify maintenance supervision, QA, and the squadron safety offices/NCO of any suspected overexposure to RF radiation.

A15.5.3. Report all suspected overexposure to RF radiation to the Installation Radiation Safety Officer (IRSO), typically a member of the Bioenvironmental Engineer (BE) flight, for investigation and documentation. If practical, leave equipment in its current physical configuration to aid BE in conducting an accurate exposure assessment. Public Health will also document the results of the investigation.

**A15.6. Astroinertial Instrument (AI) handling.** AI contains radioactive isotope Carbon-14. Personnel will not intentionally break sealed integrity of the unit. If AI unit seal is inadvertently broken, take the following actions:

A15.6.1. Close off area to ensure control.

A15.6.2. Notify Bio Environmental, Base RSO, and Base Civil Engineering Readiness and Emergency Management Flight (CES/CEX).

## Attachment 16

## 36 MXG FORM 140, CTK INVENTORY AND CONTROL LOG

**Figure A16.1. CTK Inventory and Control Log.**

| CTK INVENTORY AND CONTROL LOG | | | | CTK NO. | |
|---|---|---|---|---|---|
| DATE | EMPLOYEE NUMBER | TOOL NUMBER AND NOMENCLATURE | CTK/TOOL DESTINATION | OUT TIME/SIGNATURE | IN TIME/SIGNATURE |
| | | | | | |
| | | | | | |
| | | | | |

## Attachment 17

## 36 WG AF145, LOST TOOL/OBJECT REPORT

**Figure A17.1. Lost Tool/Object Report.**

| LOST TOOL/OBJECT REPORT | | | PAGE OF PAGES | |
|---|---|---|---|---|
| BASE | AIRCRAFT SERIAL NUMBER | ORGANIZATION AND W/C | DATE(S) | TIME LOST |
| COMPOSITE TOOL KIT NUMBER | DESCRIPTION OF OBJECT | AFTO 781A<br>PAGE BLOCK | AIRCRAFT IMPOUNDED: ☐ YES ☐ NO<br>TIME<br>DATE | |

| W/C SUPV NOTIFIED (Last Name, Initial) NAME / TIME / DATE | MAINT SUPV NOTIFIED (Last Name, Initial) NAME / TIME / DATE | MOC NOTIFIED (Last Name, Initial) NAME / TIME / DATE | CTK CUSTODIAN NOTIFIED (Last Name, Initial) NAME / TIME / DATE | QA NOTIFIED (Last Name, Initial) NAME / TIME / DATE |
|---|---|---|---|---|
| | | | | |

AREA ITEM LOST

| ITEM NO. | MAINTENANCE SUPERVISOR/SUPERINTENDENT WILL ENSURE ALL ITEMS ARE COMPLETED | INDIVIDUAL INITIALS | TIME/DATE INITIALED |
|---|---|---|---|
| 1 | NAME OF INDIVIDUAL(S) THAT LOST OBJECT | | |
| 2 | WORK BEING PERFORMED WHEN ITEM LOST | | |
| 3 | AREAS CHECKED AND STEPS TAKEN TO FIND OBJECT | | |
| 4 | SEARCH CONDUCTED BY *(List all individuals)* | | |
| 5 | ITEM WAS/WAS NOT FOUND *(If found, give location)* | | |
| 6 | COPY OF FORM FILED BY CTK CUSTODIAN | | |
| 7 | FORM FILED BY RECORDS CLERK *(If Applicable)* | | |
| 8 | SIGNATURE OF MAINTENANCE SUPERVISOR/SUPERINTENDENT | | |

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